

Azure Log Analytics KQL App - Deployment & Operations Guide

This document describes how to set up, deploy, and maintain the application in local development (Windows 11) and production (Docker, Azure hosting) environments.

1. Environment Variable Reference

Create a `.env` file in the root workspace folder. The application loads this file automatically in development and when run via Docker container.

Variable Name	Type	Default	Description
<code>NODE_ENV</code>	String	<code>development</code>	Set to <code>production</code> in containerized/production setups.
<code>PORT</code>	Number	<code>8080</code>	Port the API server listens on.
<code>CORS_ORIGIN</code>	String	<code>http://localhost:5010</code>	Allowed CORS origins (comma-separated list for multiples).
<code>LOG_ANALYTICS_WORKSPACE_ID</code>	String	<i>Required (UUID)</i>	Default Azure Log Analytics Workspace ID fallback.
<code>ALLOW_WORKSPACE_OVERRIDE</code>	Boolean	<code>false</code>	If <code>true</code> , clients can supply custom workspace IDs via API request body.
<code>QUERY_TIMEOUT_MS</code>	Number	<code>30000</code>	Query execution timeout in milliseconds.
<code>QUERY_MAX_ROWS</code>	Number	<code>5000</code>	Maximum rows returned per query (capped at <code>50000</code>).

Variable Name	Type	Default	Description
QUERY_MAX_LENGTH	Number	20000	Maximum KQL query character length limit.
RATE_LIMIT_WINDOW_MS	Number	60000	Rate limiting sliding window size in milliseconds.
RATE_LIMIT_MAX	Number	60	Max API requests allowed per client IP per window.
AZURE_TENANT_ID	String	Required for SPN	Azure AD Directory (Tenant) ID for Service Principal.
AZURE_CLIENT_ID	String	Required for SPN	Azure AD Application (Client) ID for Service Principal / Managed Identity.
AZURE_CLIENT_SECRET	String	Required for SPN	Azure AD Client Secret for Service Principal.
VITE_AZURE_CLIENT_ID	String	Required for Login	Frontend SPA Client ID for Microsoft Azure AD MSAL Login.
VITE_AZURE_TENANT_ID	String	Required for Login	Frontend SPA Tenant ID for Microsoft Azure AD MSAL Login.
VITE_REQUIRE_AZURE_AD_AUTH	Boolean	true	Set to false to bypass Azure AD login landing page and use backend credentials.
VITE_WORKSPACES	String	Optional	Comma-separated Name:GUID list of predefined Log Analytics Workspaces.

2. Where to Set Azure AD SPN Authentication & Workspace ID

A. Configuring Azure AD Service Principal (SPN) Authentication

To authenticate with Azure AD, define your Service Principal credentials and Single Page Application (SPA) IDs in `.env` :

1. In Root `.env` File (Local Dev & Docker):

```
# Backend Service Principal (SPN) Credentials
AZURE_TENANT_ID=your-tenant-id-guid
AZURE_CLIENT_ID=your-spn-client-id-guid
AZURE_CLIENT_SECRET=your-spn-client-secret-value

# Frontend Azure AD MSAL Single Page App (SPA) Login
VITE_AZURE_CLIENT_ID=your-spa-client-id-guid
VITE_AZURE_TENANT_ID=your-tenant-id-guid
```

2. In Containerized / Azure Hosting Environments (AKS / Container Apps):

- Add `AZURE_TENANT_ID` , `AZURE_CLIENT_ID` , and `AZURE_CLIENT_SECRET` as environment variables or Kubernetes secret keys in `aks/secret.yaml` .
- Set `VITE_AZURE_CLIENT_ID` and `VITE_AZURE_TENANT_ID` during the client build step or in `aks/secret.yaml` .

3. Azure Portal App Registration Setup Instructions:

1. Navigate to **Microsoft Entra ID > App Registrations > New registration**.
 2. Name: `LogAnalytics-KQL-App` .
 3. Supported Account Types: `Accounts in this organizational directory only (Single tenant)` .
 4. Redirect URI (Single-page application - SPA): `http://localhost:5010/` (for local dev) and `http://localhost:8080/` (for production).
 5. Click **Register**.
 6. Copy **Application (client) ID** into `VITE_AZURE_CLIENT_ID` and `AZURE_CLIENT_ID` .
 7. Copy **Directory (tenant) ID** into `VITE_AZURE_TENANT_ID` and `AZURE_TENANT_ID` .
 8. Go to **Certificates & secrets > New client secret** to generate a secret, and copy its value into `AZURE_CLIENT_SECRET` .
 9. Go to **API permissions > Add a permission**:
 - Select **Log Analytics API** → Delegated permissions → `Data.Read` .
 - Select **Azure Service Management** → Delegated permissions → `user_impersonation` (enables dynamic workspace discovery).
 10. Click **Grant admin consent for [Your Organization]**.
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B. Configuring Log Analytics Workspace ID

Workspace IDs can be set via two methods:

Method 1: Environment Variable Default (Server Fallback)

Set `LOG_ANALYTICS_WORKSPACE_ID` in `.env` :

```
LOG_ANALYTICS_WORKSPACE_ID=11111111-1111-1111-1111-111111111111
```

This ID serves as the default workspace when queries are executed without specifying a workspace ID.

Method 2: Dynamic UI Selection via Azure AD Login (Recommended)

1. Launch the application. Users land on the **Microsoft Azure AD Login Page**.
 2. Click **Sign in with Microsoft Azure AD**.
 3. Upon authentication, the application automatically queries Azure Resource Graph (`microsoft.operationalinsights/workspaces`) using the logged-in user's token.
 4. Select any accessible Log Analytics Workspace directly from the **Log Analytics Workspace** dropdown selector in the top bar.
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3. Local Development (Windows 11)

Prerequisites

- Node.js v20 or v22 (recommended).
- Azure CLI (`az`) installed and logged in (required for local token capture if using `DefaultAzureCredential`).

Configuration & Dev Execution

1. **Log in to Azure CLI:** Open PowerShell or CMD:

```
az login
```

Set active subscription:

```
az account set --subscription "<your-subscription-id>"
```

2. **Initialize Environment:** Copy `.env.example` to `.env` :

```
copy .env.example .env
```

Configure `AZURE_TENANT_ID` , `AZURE_CLIENT_ID` , `AZURE_CLIENT_SECRET` , `VITE_AZURE_CLIENT_ID` , `VITE_AZURE_TENANT_ID` , and `LOG_ANALYTICS_WORKSPACE_ID` .

3. **Install Dependencies:**

```
npm run install:all
```

4. **Start Dev Servers:**

```
npm run dev
```

- Frontend (Vite SPA): `http://localhost:5010`
 - Backend (Express API): `http://localhost:8080`
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4. Container Deployment (Docker)

The application includes an optimized multi-stage `Dockerfile` which builds the React frontend assets and serves them via Node.js Express.

Build and Run with Docker Compose

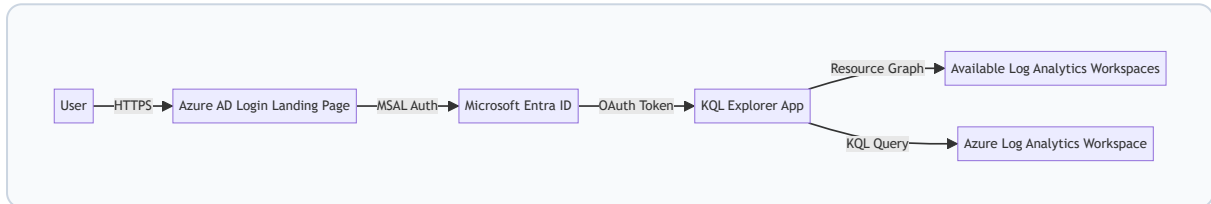
1. Verify `.env` contains `NODE_ENV=production` and valid Azure AD SPN credentials.
2. Build and start containers:

```
docker compose up --build
```

Access the app at <http://localhost:8080/>.

5. Azure Hosting & Security Configuration

For production hosting in Azure, **Azure Kubernetes Service (AKS)** or **Azure Container Apps** is recommended.



Managed Identity Setup (Alternative to SPN)

If hosting on Azure Container Apps or AKS:

1. Enable **Managed Identity** on your Azure Container resource.
2. Under the Log Analytics Workspace, navigate to **Access Control (IAM)**.
3. Click **Add role assignment** → select **Log Analytics Reader** role.
4. Assign access to the Managed Identity of your App.
5. If using User-Assigned Managed Identity, set `AZURE_CLIENT_ID` to the Client ID of the User-Assigned Identity.

6. Custom DNS Domain Setup Guide

When configuring the application to run under a custom DNS domain (e.g., `https://loganalytics.yourcompany.com` or `http://kql-app.local:5010`), follow these configuration steps:

1. Update Environment Variables (`.env`)

Update `CORS_ORIGIN` to include your custom domain:

```
CORS_ORIGIN=https://loganalytics.yourcompany.com,http://localhost:5010,http://local
```

2. Add Redirect URI in Azure AD Portal (Microsoft Entra ID)

1. Go to **Azure Portal > Microsoft Entra ID > App Registrations > [Your App Registration]**.
2. Click **Authentication** in the left sidebar.
3. Under **Single-page application (SPA)**, add your custom domain redirect URI:
 - `https://loganalytics.yourcompany.com/` (or `http://kql-app.local:5010/`)
4. Click **Save**.

3. Client MSAL Redirect Configuration (`client/src/authConfig.ts`)

The client app uses `redirectUri: "/"` by default, which automatically uses your current browser origin (`window.location.origin`). No code changes are required unless you wish to specify an explicit environment override.

4. Local Vite Dev Server Host Binding (`client/vite.config.ts`)

If mapping a custom local domain via your `hosts` file (e.g., `127.0.0.1 loganalytics.local`), enable host binding in `client/vite.config.ts`:

```
export default defineConfig({
  plugins: [react()],
  server: {
    port: 5010,
    host: true,
    allowedHosts: ["loganalytics.local", "loganalytics.yourcompany.com"],
    proxy: {
      "/api": "http://127.0.0.1:8080"
    }
  }
});
```


5. AKS / Kubernetes Ingress Configuration (`aks/istio-ingress.yaml` & `aks/secret.yaml`)

- In `aks/istio-ingress.yaml` , set the `hosts` entry to match your custom DNS domain:

```
hosts:
  - "loganalytics.yourcompany.com"
```

- In `aks/secret.yaml` , update the `CORS_ORIGIN` secret string to `https://loganalytics.yourcompany.com` .
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7. Production Deployment Setup Guide

For containerized production deployments (Docker, Azure Container Apps, or Azure Kubernetes Service - AKS):

A. Kubernetes Secret Configuration (`aks/secret.yaml`)

Set the following keys in your production environment or Kubernetes secret:

```
stringData:
  NODE_ENV: "production"
  PORT: "8080"
  CORS_ORIGIN: "https://loganalytics.yourcompany.com"

  # Log Analytics Workspace Settings
  LOG_ANALYTICS_WORKSPACE_ID: "11111111-1111-1111-1111-111111111111"
  ALLOW_WORKSPACE_OVERRIDE: "true"
  VITE_WORKSPACES: "Prod-App-Logs:11111111-1111-1111-1111-111111111111,Prod-Security:11111111-1111-1111-1111-111111111111"

  # Authentication Mode Switch ("false" to bypass login landing screen in production)
  VITE_REQUIRE_AZURE_AD_AUTH: "false"

  # Azure Backend Authentication (Service Principal or Managed Identity)
  AZURE_TENANT_ID: "<production-tenant-id>"
  AZURE_CLIENT_ID: "<production-spn-client-id>"
  AZURE_CLIENT_SECRET: "<production-spn-client-secret>"
```

B. Choosing Production Authentication Mode

- **Mode 1: Internal Portal / SPN Mode (`VITE_REQUIRE_AZURE_AD_AUTH=false`):**
 - Set `VITE_REQUIRE_AZURE_AD_AUTH=false` .
 - Users enter the query workspace immediately without seeing an Azure AD login landing screen.
 - Queries execute via the backend Service Principal (`AZURE_CLIENT_SECRET`) or container Managed Identity against the production Log Analytics Workspaces listed in `VITE_WORKSPACES` .
- **Mode 2: User Single Sign-On (`VITE_REQUIRE_AZURE_AD_AUTH=true`):**
 - Set `VITE_REQUIRE_AZURE_AD_AUTH=true` .
 - Set `VITE_AZURE_CLIENT_ID` & `VITE_AZURE_TENANT_ID` .
 - Ensure `https://loganalytics.yourcompany.com/` is added under **Microsoft Entra ID > App Registrations > Authentication > SPA Redirect URIs**, and click **Grant Admin Consent** in Azure Portal.

8. Summary of Recent UI & System Feature Updates

1. 🌿 Pistachio Green Theme & Design System:

- Palette: Deep Slate Pistachio background (`#091a12` / `#112d20`), Vibrant Green accents (`#4ade80`), Pistachio Lime column headers (`#a3e635`), and glowing green button gradients (`linear-gradient(90deg, #22c55e 0%, #15803d 100%)`).

2. 🖱️ Direct Click-Hold Cursor Column Movement:

- Output table columns are reordered directly by clicking, holding, and dragging headers (`cursor: grab` / `cursor: grabbing`) without requiring manual arrow buttons (`◀` / `▶`).

3. 🔗 Hierarchical Dependent Dynamic Filters:

- **AFD Logs:** `Resource` → `hostName_s` / `host_s` (DNS Name).
- **App Gateway Logs:** `Resource` → `originalHost_s` (Original Host).
- **Storage Logs:** `AccountName` → `FileShareName` / `ContainerName` .
- **Firewall Logs:** `Resource` .
- Dynamic filters include inline search boxes (`.filter-search-field`) and name-wise alphabetical sorting.

4. 📐 50% Width Workspace Selector Container:

- Workspace selection tab is constrained to 50% screen width (`width: 50%; max-width: 50%`) with unified `'Inter'` typography across all UI dropdowns and labels.